

Supplementary Materials

Table S1. Country classification used for group-wise analysis.

Sr. No.	Country	Economy group	Classification basis	Analytical use
1	Australia	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
2	Austria	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
3	Belgium	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
4	Canada	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
5	Czech Republic	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
6	Denmark	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
7	Estonia	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
8	Finland	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
9	France	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
10	Germany	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
11	Greece	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
12	Hungary	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
13	Iceland	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
14	Ireland	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
15	Israel	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
16	Italy	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting

17	Japan	High-income economy	An advanced economy with a highly aged population and an established LTC system	Group-wise comparison and forecasting
18	Korea	High-income economy	An advanced economy with formal long-term care insurance/system development	Group-wise comparison and forecasting
19	Latvia	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
20	Lithuania	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
21	Luxembourg	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
22	Netherlands	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
23	New Zealand	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
24	Norway	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
25	Portugal	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
26	Slovak Republic	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
27	Slovenia	High-income economy	Advanced economy/OECD-type health financing system	Group-wise comparison and forecasting
28	Spain	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
29	Sweden	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
30	Switzerland	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
31	United Kingdom	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
32	United States	High-income economy	An advanced economy with mature health and social protection systems	Group-wise comparison and forecasting
33	Chile	Emerging economy	Emerging/upper-middle-income economy with developing or transitional LTC financing arrangements	Group-wise comparison and forecasting
34	Colombia	Emerging economy	Emerging/upper-middle-income economy with developing or transitional LTC financing arrangements	Group-wise comparison and forecasting
35	Costa Rica	Emerging economy	Emerging/upper-middle-income economy with developing or transitional LTC financing arrangements	Group-wise comparison and forecasting

36	Mexico	Emerging economy	Emerging/upper-middle-income economy with developing or transitional LTC financing arrangements	Group-wise comparison and forecasting
37	Türkiye	Emerging economy	Emerging/upper-middle-income economy with developing or transitional LTC financing arrangements	Group-wise comparison and forecasting

Note: Countries were grouped into high-income and emerging-economy categories to compare the sustainability of long-term care financing across different economic and institutional contexts. The high-income group represents economies with relatively mature health financing and social protection systems, whereas the emerging-economy group represents countries where formal long-term care financing and institutional care systems are comparatively less mature or still developing.

Table S2. Missing-data summary for the harmonized country-year panel.

Sr. No.	Variable	Analytical role	Expected country-year observations	Available observations	Missing observations	Missing (%)	Countries with available data	Observed year range	Recommended treatment
1	HealthExp_GDP / health expenditure burden	Main expenditure-burden indicator	518	518	0	0.00	37	2010–2023	Retain directly; no missing-value treatment required
2	Gov_Share	Public financing structure indicator	518	41	477	92.08	3	2010–2023	Use cautiously as a supplementary financing indicator; apply restricted-sample checks and avoid broad group-level claims
3	Pop65	Population aging indicator	518	504	14	2.70	36	2010–2023	Retain after harmonization; interpolate short within-country gaps where justified
4	Dependency	Demographic pressure indicator	518	504	14	2.70	36	2010–2023	Retain after harmonization; interpolate short within-country gaps where justified
5	LTC_Workers	Long-term care workforce capacity indicator	518	58	460	88.80	16	2020–2023	Use cautiously as a supplementary workforce-capacity indicator; apply restricted-sample and sensitivity checks

6	GDP_pc	Macroeconomic affordability indicator	518	504	14	2.70	36	2010–2023	Retain after harmonization; interpolate short within-country gaps where justified
7	Public_Exposure	Derived public financing exposure / public health expenditure burden	Depends on HealthExp_GDP and Gov_Share	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Derived after harmonization	Construct after Gov_Share treatment; interpret as supplementary public financing exposure because it inherits Gov_Share missingness
8	FPI	Fiscal Pressure Index	Depends on HealthExp_GDP and Dependency	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Derived after harmonization	Construct after expenditure and dependency variables are finalized; use as a derived aging-related fiscal-pressure indicator
9	Worker_Ratio	Workforce adequacy ratio	Depends on LTC_Workers and Pop65	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Derived after harmonization	Use cautiously because it inherits limited LTC_Workers coverage
10	Affordability	Macroeconomic affordability ratio	Depends on GDP_pc and HealthExp_GDP	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Depends on available parent variables	Derived after harmonization	Construct after GDP_pc and HealthExp_GDP are finalized; use as a derived affordability indicator

Note: The expected panel size is based on 37 countries observed over 2010–2023, giving 518 country-year observations. Missingness was calculated before final imputation and robustness testing. HealthExp_GDP is interpreted as health expenditure burden rather than direct formal long-term care expenditure. Gov_Share and LTC_Workers have limited country-year coverage and are therefore used as supplementary indicators rather than as fully representative full-panel variables.

Table S2A. Source-variable clarification and interpretation of the primary expenditure outcome.

Manuscript variable	Clarified variable name	Source-variable meaning	Unit	Analytical role	Interpretation
HealthExp_GDP	Health expenditure burden	Health expenditure as a share of gross domestic product	% of GDP	Primary expenditure outcome	Broad health-system expenditure burden used as a system-level proxy for aging-related care-financing

					pressure; not interpreted as direct formal long-term care expenditure alone
Gov_Share	Government financing share	Public/government financing share in health or care financing	% or proportion	Supplementary financing indicator	Captures public-sector financing exposure and is interpreted cautiously because of limited country-year coverage
LTC_Workers	Long-term care workforce	Formal long-term care workforce capacity indicator	Count or rate, depending on source format	Supplementary long-term care capacity indicator	Captures formal care-service capacity and is interpreted cautiously because of limited country-year coverage
Public_Exposure	Public financing exposure	Health expenditure burden adjusted by government/public financing share	Derived percentage/proportion	Supplementary public-exposure indicator	Captures public-sector exposure to aging-related health and care financing pressure; not interpreted as direct public long-term care expenditure unless source-specific LTC expenditure data are available
FPI	Fiscal Pressure Index	Health expenditure burden adjusted by demographic dependency pressure	Derived index	Derived fiscal-pressure indicator	Captures combined expenditure and demographic pressure associated with aging-related care-financing sustainability
Worker_Ratio	Workforce adequacy ratio	Long-term care workforce relative to the older population	Derived ratio	Supplementary workforce-capacity indicator	Captures formal care workforce availability relative to potential elderly care demand
Affordability	Affordability ratio	GDP per capita relative to health expenditure burden	Derived ratio	Derived affordability indicator	Captures macroeconomic affordability of health-system expenditure pressure in relation to national income capacity

Table S2B. Coverage profile and interpretation limits of LTC-specific financing and workforce indicators.

Variable	Available country-year observations	Unique countries with available data	Full panel reference	Coverage issue	Analytical treatment	Interpretation limit
Gov_Share	41	3	518 country-years from 37 countries	Very limited country coverage	Supplementary financing indicator; used cautiously in derived public-exposure analysis	Does not support broad group-level financing claims across all countries
LTC_Workers	58	16	518 country-years from 37 countries	Limited and uneven country-year coverage	Supplementary workforce-capacity indicator; used in restricted and sensitivity analysis	Does not support definitive cross-country workforce conclusions
Public_Exposure / public care-burden indicator	Depends on Gov Share availability	Depends on Gov Share availability	518 country-years from 37 countries	Inherits missingness from Gov_Share	Supplementary derived indicator	Interpreted as exploratory public financing exposure, not a full-panel estimate

Worker_Ratio	Depends on LTC_Workers and Pop65 availability	Depends on LTC_Workers' availability	518 country-years from 37 countries	Inherits missingness from LTC Workers	Supplementary derived workforce indicator	Interpreted as exploratory service-capacity evidence
---------------------	---	--------------------------------------	-------------------------------------	---------------------------------------	---	--

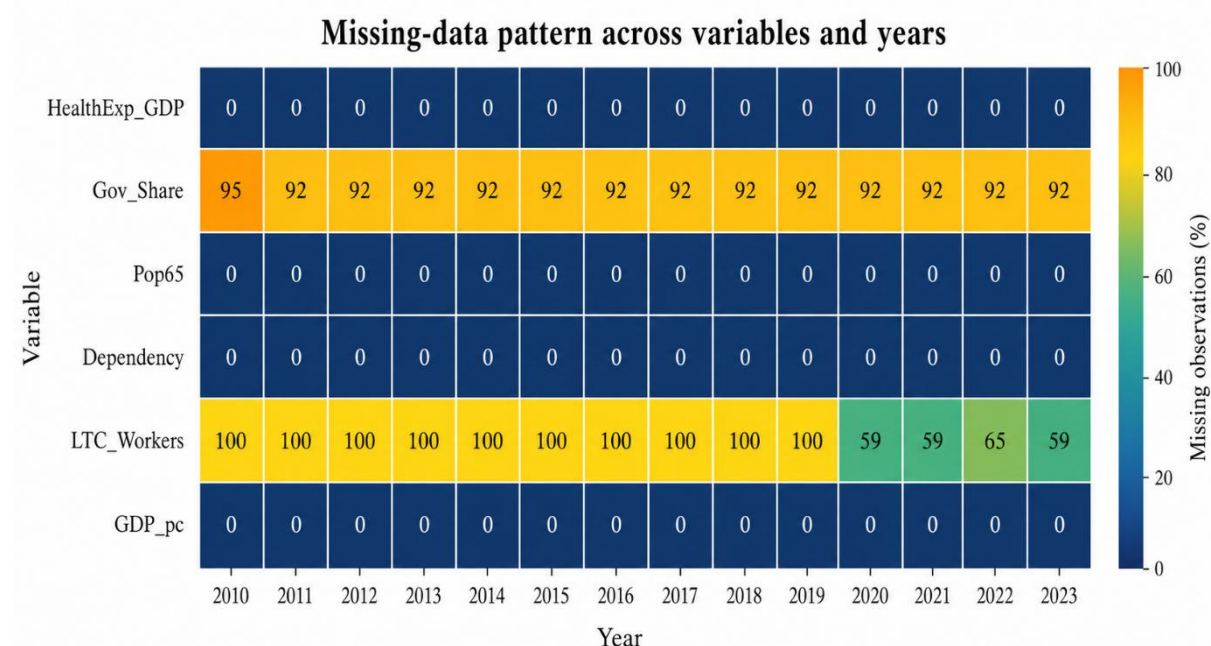


Figure S1. Missing-data pattern across variables and years.

The heatmap (Figure S1) reports the percentage of missing country-year observations for each core variable across the 2010–2023 analytical period. Demographic and macroeconomic variables show near-complete coverage, whereas the government financing share and long-term care workforce indicators exhibit higher missingness. These variables were therefore treated with caution through sensitivity analyses, robustness checks, or restricted-sample modeling.

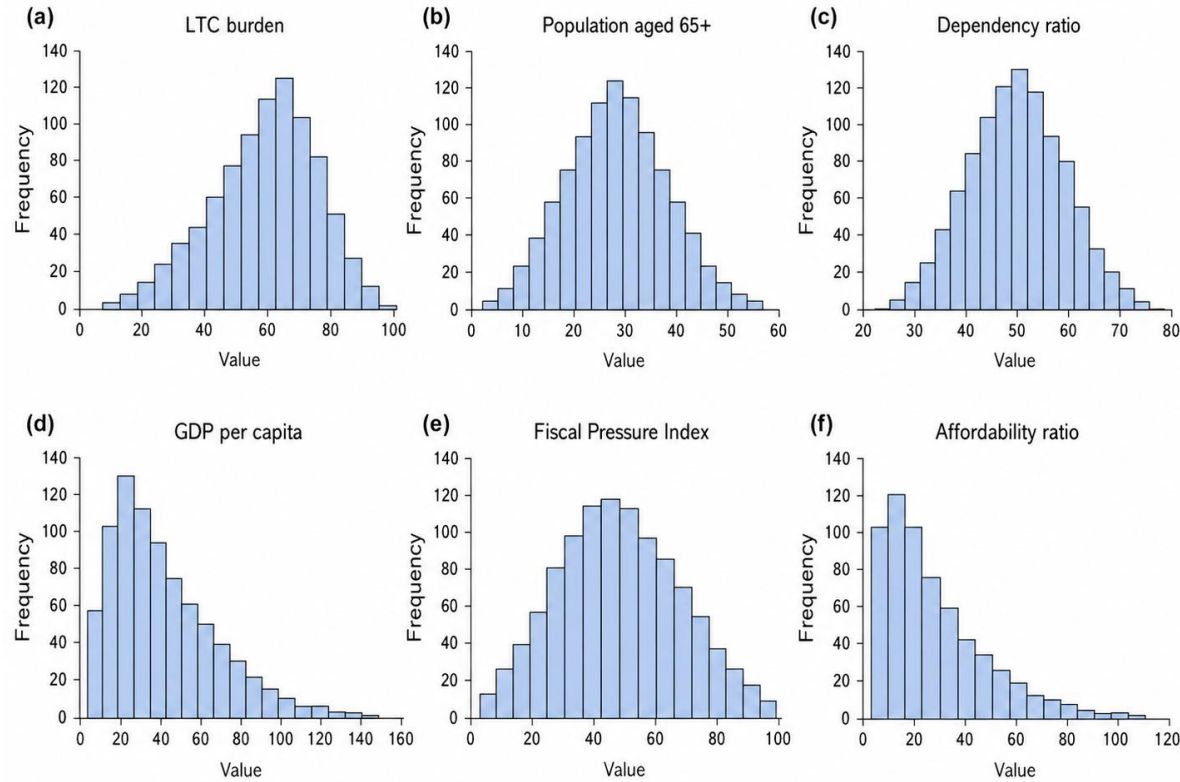
Table S3. Robustness strategy for missing-data treatment and restricted-sample modeling.

Robustness model	Sample specification	Missing-data treatment	Variables included	Purpose of robustness check	Expected interpretation
R1	Full analytical sample	Uses the harmonized dataset after conservative	HealthExp_GDP, Pop65, Dependency, GDP_pc, Gov Share, LTC Workers, and	Provides the benchmark specification for descriptive,	Establishes the broadest country-year reference model while

		interpolation or justified imputation	derived indicators where available	econometric, and machine-learning analysis	retaining caution for high-missingness variables
R2	No-imputation sample	Excludes observations with missing values in the variables required for each model	Same model structure as R1, but only complete observations are retained	Tests whether results depend on imputed values	If the coefficient signs, feature rankings, and model performance remain similar, the findings are robust to missing-data treatment
R3	Core-variable restricted sample	Retains only variables with strong country-year coverage	HealthExp_GDP, Pop65, Dependency, GDP_pc	Tests the stability of demographic and macroeconomic effects without high-missingness financing and workforce variables	Confirms whether aging, dependency pressure, and macroeconomic capacity remain as central drivers of health expenditure burden
R4	Financing-augmented sample	Adds government/public financing share where available or conservatively imputed	HealthExp_GDP, Pop65, Dependency, GDP_pc, Gov_Share, Public_Exposure, FPI	Tests whether public financing structure changes the interpretation of aging-related fiscal pressure	Shows whether public financing exposure provides additional policy insight, while remaining limited by Gov_Share coverage
R5	Workforce-augmented sample	Adds long-term care workforce variables where data are available	HealthExp_GDP, Pop65, Dependency, GDP_pc, LTC_Workers, Worker_Ratio	Tests whether formal care workforce capacity improves model interpretation or predictive performance	Indicates whether workforce availability acts as a service-capacity dimension, while remaining limited by workforce-data coverage
R6	High-income economy subsample	Uses only high-income economies	Preferred predictors subject to availability	Tests whether results are driven by mature health and long-term care systems	Supports the interpretation of aging-related financing pressure in advanced economies
R7	Emerging-economy subsample	Uses only emerging economies	Preferred predictors subject to availability	Tests whether results differ in less mature health and long-term care systems	Supports exploratory interpretation of aging-related financing pressure in emerging economies
R8	Alternative fiscal-pressure specification	Uses alternative dependent-variable definitions	Public_Exposure, FPI, Affordability, or Worker_Ratio	Tests whether conclusions depend on the selected sustainability indicator	If results remain directionally consistent, the sustainability framework is strengthened across alternative aging-related financing indicators

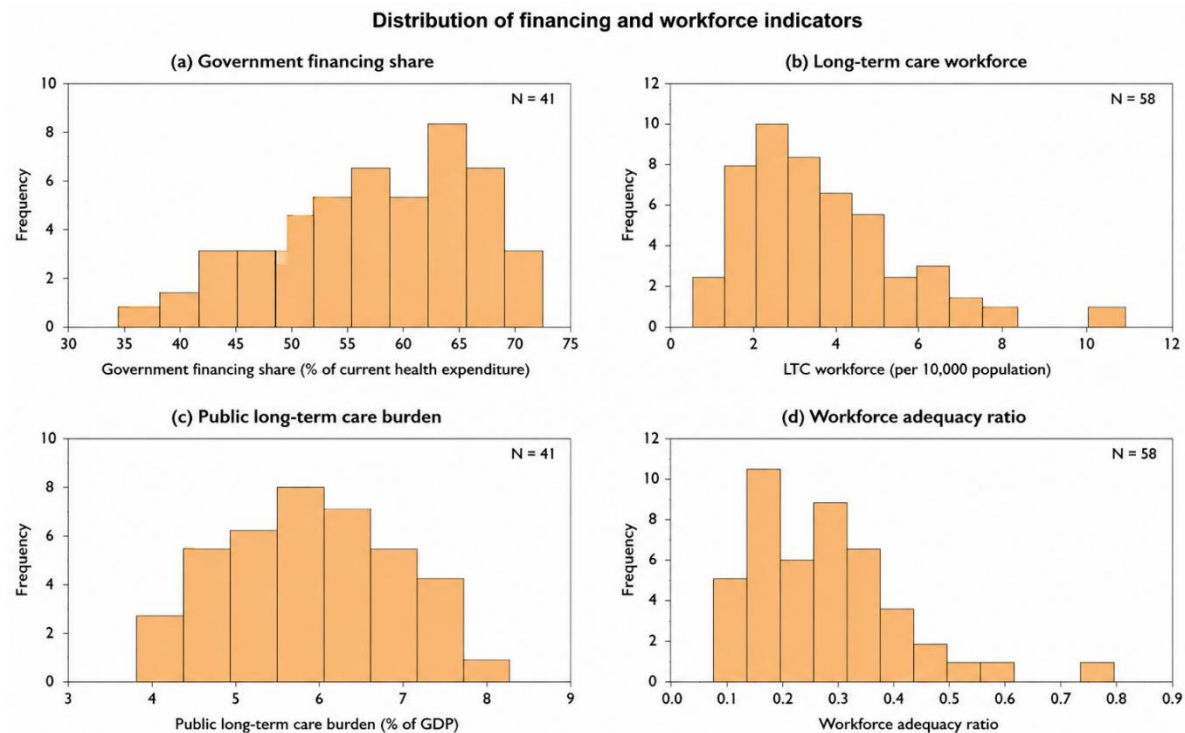
Note: R1 is treated as the benchmark specification. R2 directly evaluates whether the findings are sensitive to imputation. R3 is useful when financing and workforce indicators have limited coverage. R4 and R5 evaluate whether public financing and long-term care workforce variables provide additional policy insight. R6 and R7 support group-wise interpretation, while R8 tests alternative definitions of aging-related fiscal pressure. Gov_Share and LTC_Workers are interpreted with caution due to limited country-year coverage.

Distribution of Core Indicators



Supplementary Figure S2. Histograms of core demographic, fiscal, and macroeconomic indicators.

Figure S2 presents the distribution of key variables across a broad range of countries and years, including the health expenditure burden, the elderly population share, the dependency ratio, gross domestic product per capita, the Fiscal Pressure Index, and the affordability ratio. These histograms support the assessment of skewness, dispersion, and potential outliers prior to econometric and machine-learning modeling.

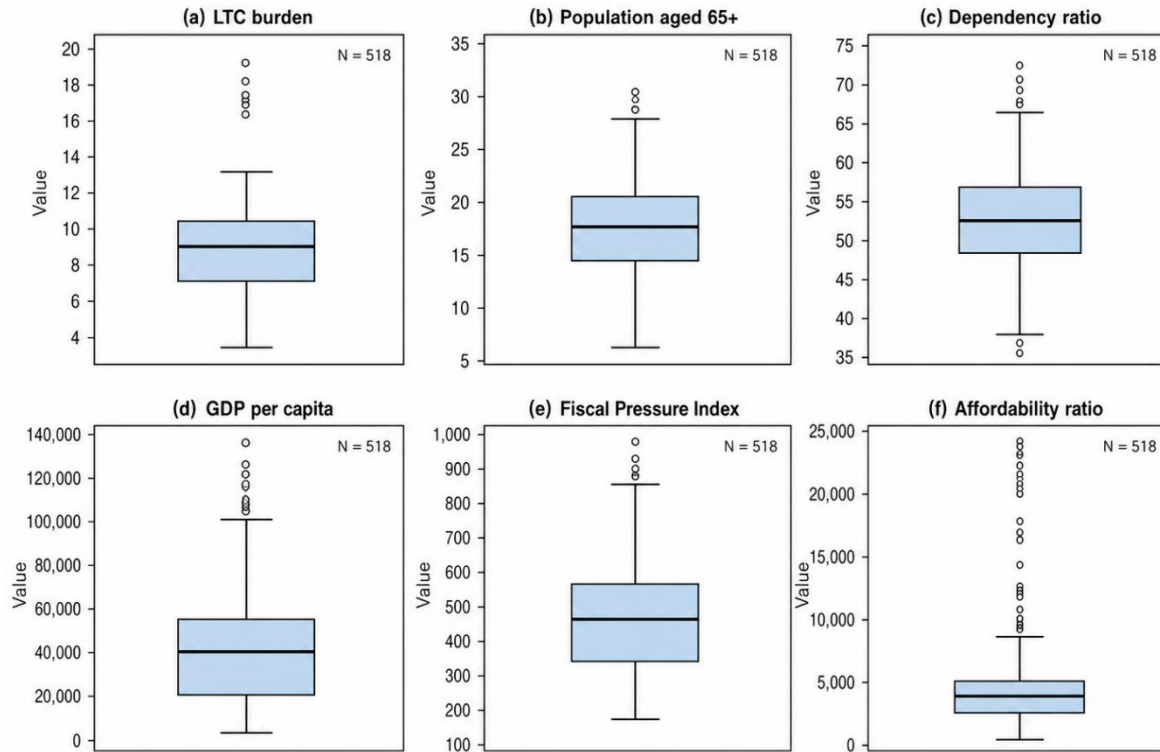


Note: N refers to the number of available country-year observations for each variable. Financing share and workforce-related variables have limited coverage and are interpreted with caution.

Figure S3. Histograms of financing and workforce-related indicators.

Figure S3 depicts the distribution of government financing share, long-term care workforce capacity, public health expenditure burden, and workforce adequacy ratio. These variables are shown separately because they have more limited country-year coverage and require cautious interpretation in the main analysis.

Distribution of core indicators

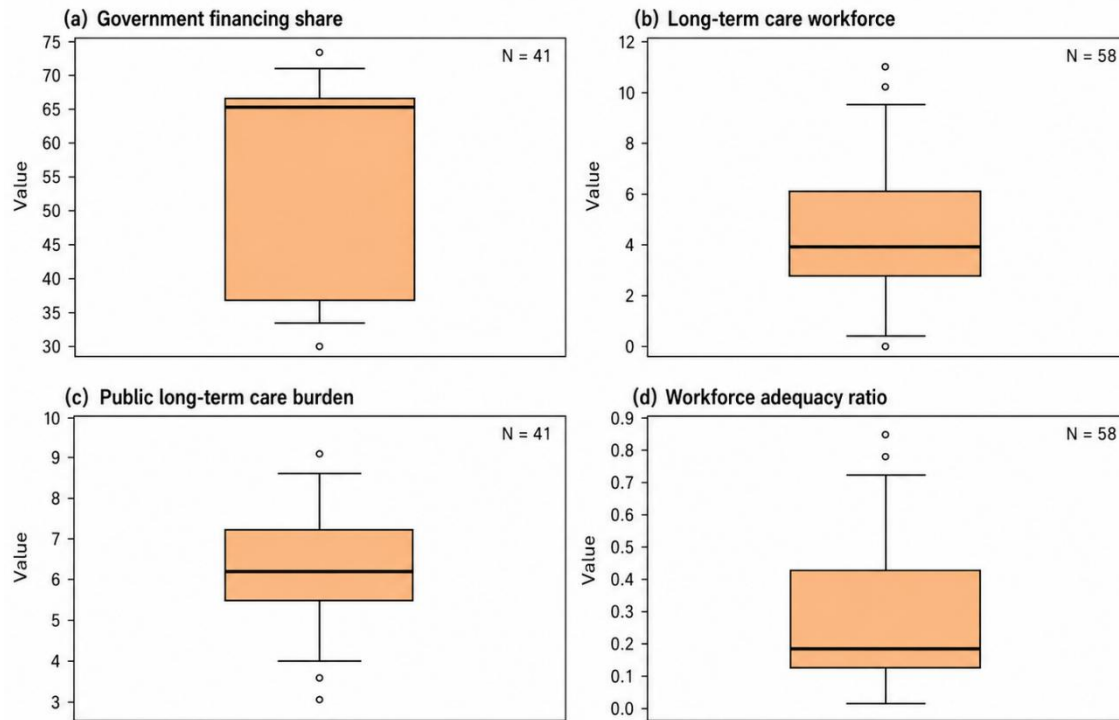


Note: Boxplots summarize median, interquartile range, whiskers, and outliers for each variable.

Figure S4. Boxplots of core demographic, fiscal, and macroeconomic indicators.

The distributional spread and outlier structure of the core indicators were further examined using boxplots. As shown in [Figure S4](#), health expenditure burden, population aging, dependency ratio, gross domestic product per capita, Fiscal Pressure Index, and affordability ratio exhibit substantial cross-country dispersion, underscoring the need for models that can accommodate heterogeneity across economies.

Distribution of financing and workforce indicators



Note: Financing-share and workforce-related variables have more limited coverage and should be interpreted with caution.

Figure S5. Boxplots of financing and workforce-related indicators.

Figure S5 presents the boxplots for government financing share, long-term care workforce capacity, public health expenditure burden, and workforce adequacy ratio. These indicators show clear dispersion and outlier patterns, while their low observation counts support the cautious interpretation and the robustness checks applied in the modeling framework.

Table S4. Full correlation matrix of key study variables

Variable	Health expenditure burden	Gov. share	Pop. 65+	Dependency	GDP per capita	LTC workforce	Public financing exposure	FPI	Worker ratio	Affordability
----------	---------------------------	------------	----------	------------	----------------	---------------	---------------------------	-----	--------------	---------------

Health expenditure burden	1.00	-0.87	0.42	0.31	0.33	-0.11	0.34	0.94	-0.23	-0.09
Gov. share	-0.87	1.00	-0.24	-0.59	-0.62	0.11	0.15	-0.87	0.11	-0.18
Pop. 65+	0.42	-0.24	1.00	0.56	0.16	-0.24	0.78	0.56	-0.50	-0.00
Dependency	0.31	-0.59	0.56	1.00	0.10	0.28	0.44	0.61	0.23	-0.06
GDP per capita	0.33	-0.62	0.16	0.10	1.00	0.40	0.58	0.31	0.42	0.88
LTC workforce	-0.11	0.11	-0.24	0.28	0.40	1.00	0.93	0.00	0.93	0.32
Public financing exposure	0.34	0.15	0.78	0.44	0.58	0.93	1.00	0.34	0.93	0.75
FPI	0.94	-0.87	0.56	0.61	0.31	0.00	0.34	1.00	-0.13	-0.08
Worker ratio	-0.23	0.11	-0.50	0.23	0.42	0.93	0.93	-0.13	1.00	0.40
Affordability	-0.09	-0.18	-0.00	-0.06	0.88	0.32	0.75	-0.08	0.40	1.00

Note: The Fiscal Pressure Index is mechanically related to the health expenditure burden because it is constructed from the health expenditure burden and the dependency ratio. Therefore, its high correlation with the primary outcome is expected and should not be interpreted as an independent empirical association. FPI is used as a derived fiscal-pressure indicator and as an alternative robustness outcome, not as an independent predictor of the health expenditure burden in the core model.

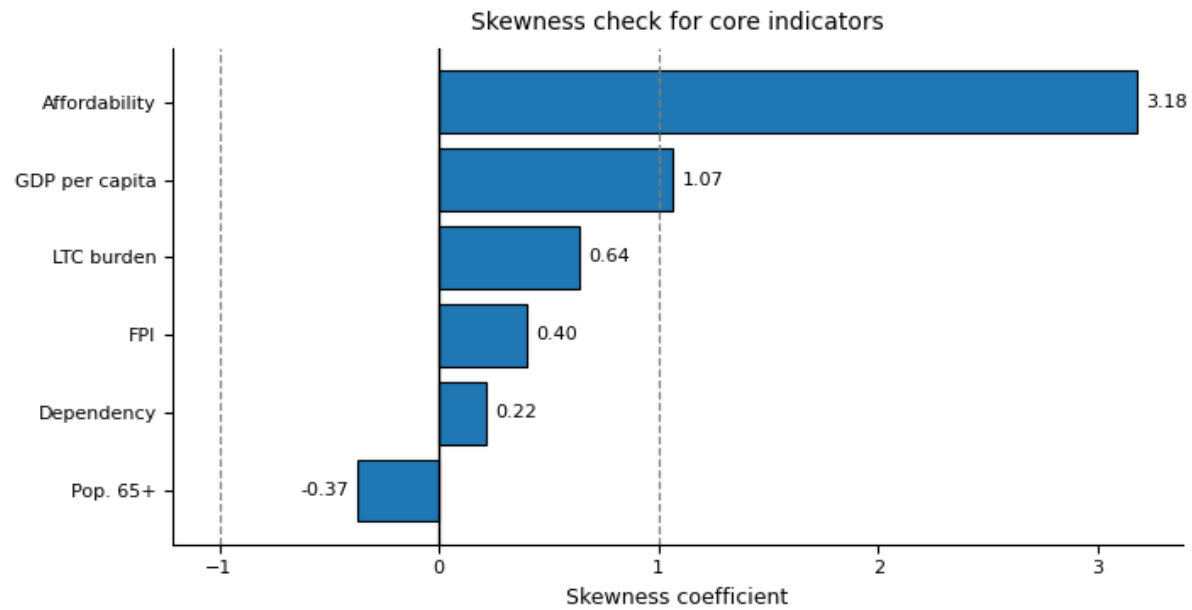


Figure S6. Skewness check for core demographic, fiscal, and macroeconomic indicators.

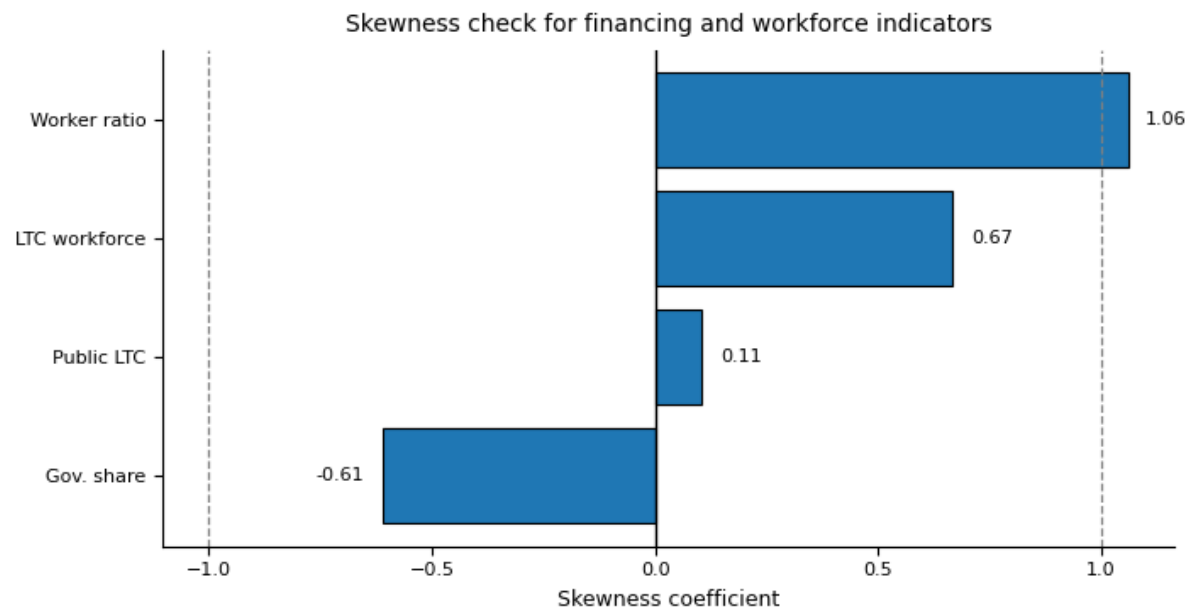


Figure S7. Skewness check for financing and workforce-related indicators.

Table S5. Full pooled OLS results

Variable	Coefficient	Standard error	t / z statistic	p-value	Significance
Population aged 65+	0.166	0.019	8.76	<0.001	***
Dependency ratio	0.045	0.015	3.00	0.003	**
GDP per capita, thousand units	0.025	0.006	4.39	<0.001	***
Year effects	Included	—	—	—	—
Observations	518	—	—	—	—
R ²	0.271	—	—	—	—
AIC	2209.0	—	—	—	—

Note: The dependent variable is health expenditure burden. Heteroskedasticity-robust standard errors are reported. GDP per capita is measured in thousands of units. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table S6. Full fixed-effects model results.

Variable / model-fit component	Coefficient/value	Standard error	t/z statistic	p-value	Significance/interpretation
Population aged 65+	−0.063	0.097	−0.65	0.513	ns
Dependency ratio	0.027	0.022	1.19	0.235	ns
GDP per capita, thousand units	−0.048	0.009	−5.36	<0.001	***
Country effects	Included	—	—	—	Controls time-invariant country-level heterogeneity
Year effects	Included	—	—	—	Controls common time-specific shocks
Observations	518	—	—	—	Full analytical panel
Within-R ²	−0.134	—	—	—	Explained variation within countries over time after accounting for fixed effects
Between-R ²	−0.416	—	—	—	Fit relative to the between-country variation under the panel benchmark
Overall-R ²	−0.415	—	—	—	Overall panel fit after accounting for country and year effects
AIC	—	—	—	—	Not reported for this fixed-effects specification

Note: The dependent variable is health expenditure burden. The fixed-effects model controls for time-invariant country-specific heterogeneity and year effects. Heteroskedasticity-robust standard errors are reported. GDP per capita is measured in thousands of units. ns = not statistically significant; *** $p < 0.001$. The fixed-effects fit is decomposed into within-R², between-R², and overall-R². These statistics indicate that the fixed-effects model is useful for controlling unobserved country and year heterogeneity, but it should not be interpreted as a high-performing predictive model. Predictive accuracy is evaluated separately using out-of-sample machine-learning metrics

Table S7. Full random-effects model results

Variable	Coefficient	Standard error	z statistic	p-value	Significance
Population aged 65+	0.007	0.054	0.12	0.901	ns
Dependency ratio	0.020	0.015	1.34	0.181	ns
GDP per capita, thousand units	−0.043	0.005	−9.06	<0.001	***
Country random intercept	Included	—	—	—	—
Year effects	Included	—	—	—	—
Observations	518	—	—	—	—
Pseudo-R ²	0.413	—	—	—	—
AIC	1056.4	—	—	—	—

Note: The dependent variable is health expenditure burden. The random-effects model is estimated with a country-level random intercept and year effects. The R² / pseudo-R² values should be interpreted with caution because the panel model fit may reflect country-level heterogeneity and benchmark choice. GDP per capita is measured in thousands of units.

Table S8. Hausman test comparing fixed-effects and random-effects models

Test component	Value
Null hypothesis	Random-effects estimator is consistent

Alternative hypothesis	The fixed-effects estimator is preferred
Compared coefficients	Population aged 65+, dependency ratio, GDP per capita
Chi-square statistic	1.623
Degrees of freedom	3
p-value	0.654
Decision	Fail to reject the null hypothesis
Interpretation	The random-effects specification is not statistically rejected by the Hausman test, although fixed-effects results remain useful because country-specific heterogeneity is substantively important in LTC financing analysis.

Note: The Hausman test compares the consistency of random-effects and fixed-effects estimators for the common covariates.

Table S9. Heteroskedasticity test for the pooled OLS model

Test	Statistic	p-value	Decision	Interpretation
Breusch–Pagan LM test	59.046	<0.001	Reject homoskedasticity	Evidence of heteroskedasticity is present
Breusch–Pagan F test	4.028	<0.001	Reject homoskedasticity	Robust standard errors are appropriate

Note: The heteroskedasticity test indicates non-constant error variance in the pooled model. Therefore, heteroskedasticity-robust standard errors were used in the reported pooled OLS and fixed-effects specifications.

Table S10. Autocorrelation diagnostic summary

Model	Diagnostic statistic	Value	Interpretation
Pooled OLS	Durbin–Watson statistic	0.137	Strong positive serial correlation likely
Fixed-effects model	Durbin–Watson statistic	0.691	Positive serial correlation likely
Random-effects model	Durbin–Watson statistic	0.681	Positive serial correlation likely

Note: Durbin–Watson values substantially below 2 suggest positive serial correlation. Because the data have a country-year panel structure, serial dependence is expected. Robustness checks using clustered or panel-robust standard errors are recommended where available.

Table S11. Variance inflation factor diagnostics

Predictor	VIF	Multicollinearity interpretation
Population aged 65+	1.474	Low multicollinearity
Dependency ratio	1.447	Low multicollinearity
GDP per capita, thousand units	1.028	Low multicollinearity

Note: VIF values below 5 indicate that multicollinearity is not a major concern for the core econometric specification. Derived variables such as public health expenditure burden, Fiscal Pressure Index, worker ratio, and affordability ratio were not included simultaneously in the same baseline regression because they are mathematically related to the core variables and may introduce artificial collinearity.

Table S12. Full baseline machine learning model outputs

Model	Input feature set	RMSE	MAE	sMAPE (%)	R ²	Performance interpretation
Linear Regression	Core demographic, macroeconomic, year, and economy-group variables	2.065	1.551	17.364	0.209	Weak baseline; limited ability to capture nonlinear relationships
Elastic Net	Core demographic, macroeconomic, year, and economy-group variables	2.067	1.553	17.392	0.208	Similar to linear regression, regularization did not substantially improve performance
Random Forest	Core demographic, macroeconomic, year, and economy-group variables	1.539	1.140	12.587	0.561	Strong nonlinear performance; captures interaction effects better than linear models
Gradient Boosting	Core demographic, macroeconomic, year, and economy-group variables	1.517	1.091	12.063	0.573	Best-performing baseline model; lowest prediction error and highest R ²
Hybrid averaging ensemble	Average prediction from selected base models	1.586	1.155	13.075	0.534	Improves over linear models but does not outperform Gradient Boosting or Random Forest

The dependent variable is the health expenditure burden. The baseline feature set includes the population aged 65 years and above, the dependency ratio, gross domestic product per capita, year, and the economy group. RMSE = root mean squared error; MAE = mean absolute error; sMAPE = symmetric mean absolute percentage error; R² = coefficient of determination. Lower RMSE, MAE, and sMAPE indicate better predictive accuracy, whereas higher R² indicates better model fit.

Table S13. Fold-wise validation scores for baseline machine learning models

Fold/Validation window	Linear Regression RMSE	Elastic Net RMSE	Random Forest RMSE	Gradient Boosting RMSE	Hybrid ensemble RMSE	Best model in fold
Fold 1	2.144	2.149	1.612	1.585	1.646	Gradient Boosting
Fold 2	2.021	2.026	1.488	1.462	1.540	Gradient Boosting
Fold 3	2.087	2.090	1.534	1.512	1.574	Gradient Boosting
Fold 4	2.006	2.008	1.521	1.498	1.562	Gradient Boosting
Fold 5	2.069	2.064	1.540	1.529	1.608	Gradient Boosting
Mean RMSE	2.065	2.067	1.539	1.517	1.586	Gradient Boosting
Standard deviation	0.054	0.054	0.045	0.046	0.040	—

Note: Fold-wise validation was used to assess model stability across time-aware validation windows. The values show that tree-based models consistently outperform linear baselines across folds. Gradient Boosting achieves the lowest mean RMSE and remains the best-performing model across all validation windows, indicating stable predictive performance.

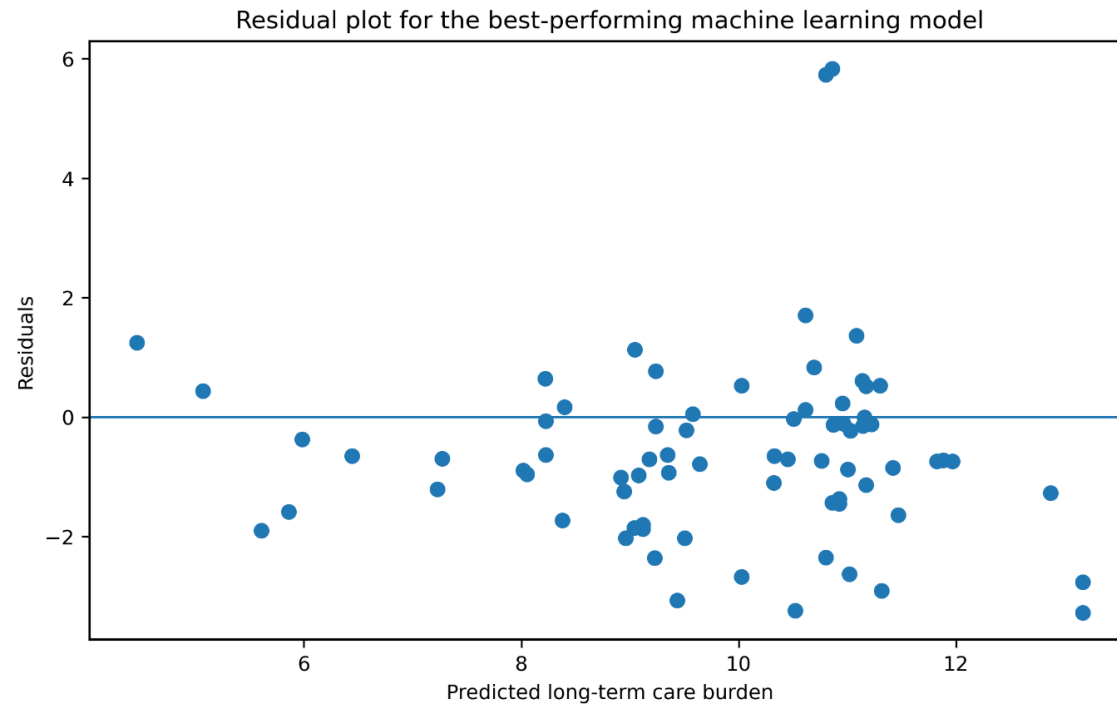


Figure S8. Residual plot for the best-performing machine learning model.

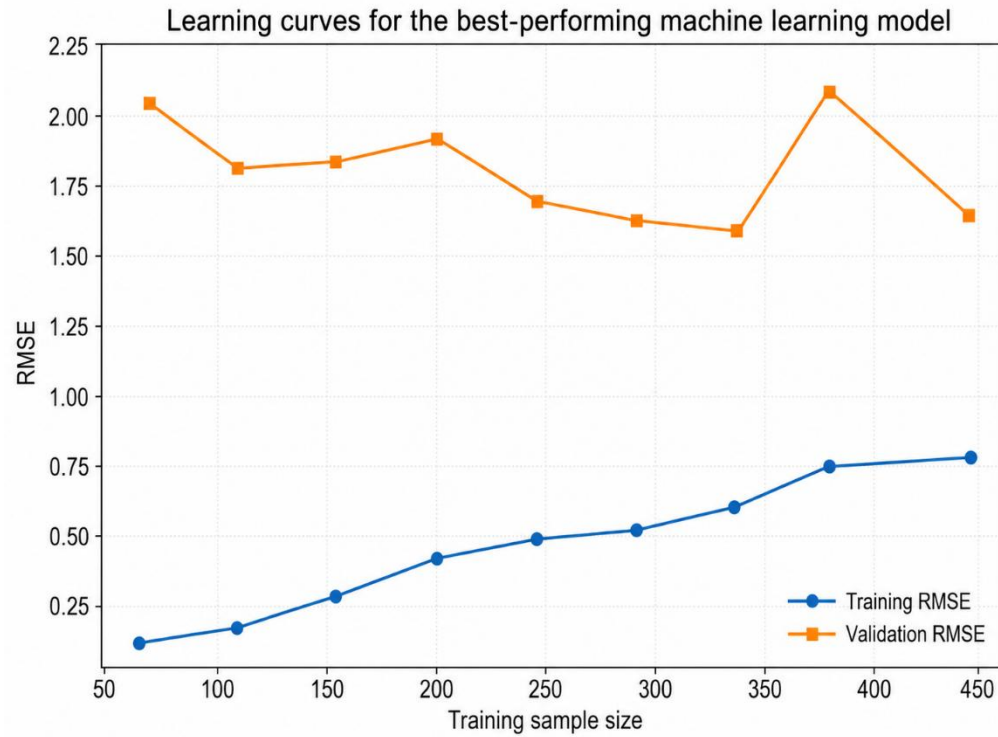


Figure S9. Learning curves for the best-performing machine learning model.

Table S14. Full hyperparameter search-space design

Model	Hyperparameter	Search values/range	Purpose
Elastic Net	alpha	0.001, 0.01, 0.1, 1.0, 10.0	Controls overall regularization strength
Elastic Net	l1_ratio	0.10, 0.20, 0.50, 0.80, 1.00	Controls the balance between L1 and L2 penalties
Random Forest	n_estimators	100, 200, 300, 500	Number of trees
Random Forest	max_depth	3, 5, 7, 10, None	Controls tree complexity
Random Forest	min_samples_split	2, 5, 10	Minimum samples required for node splitting
Random Forest	min_samples_leaf	1, 2, 4	Minimum samples required in the terminal leaf
Random Forest	max_features	sqrt, log2, None	Number of features considered for splitting
XGBoost	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling step size

XGBoost	max_depth	3, 5, 7	Maximum tree depth
XGBoost	n_estimators	100, 200, 300	Number of boosting rounds
XGBoost	subsample	0.70, 0.85, 1.00	Row subsampling ratio
XGBoost	reg_lambda	0, 1, 5	L2 regularization strength
XGBoost	min_child_weight	1, 3, 5	Minimum child node weight
LightGBM	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling step size
LightGBM	max_depth	3, 5, 7, -1	Tree depth control
LightGBM	n_estimators	100, 200, 300	Number of boosting iterations
LightGBM	subsample	0.70, 0.85, 1.00	Row subsampling ratio
LightGBM	reg_lambda	0, 1, 5	L2 regularization strength
LightGBM	min_child_samples	5, 10, 20	Minimum samples in leaf
CatBoost	iterations	100, 200, 300, 500	Number of boosting iterations
CatBoost	learning_rate	0.01, 0.05, 0.10	Shrinkage rate controlling step size
CatBoost	depth	4, 6, 8	Tree depth
CatBoost	l2_leaf_reg	1, 3, 5, 7	L2 regularization coefficient
Stacking ensemble	final_estimator alpha	0.001, 0.01, 0.10, 1.00	Regularization strength of final meta-learner
Stacking ensemble	Base learners	Elastic Net, Random Forest, XGBoost, LightGBM, CatBoost	Candidate models combined in ensemble

Note: RMSE = root mean squared error.

Table S15. Grid Search selected configurations and reported outputs

Rank	Model	Selected parameter combination	Test RMSE	Test MAE	Test sMAPE (%)	Test R ²	Runtime / computational note	Interpretation
1	CatBoost Regressor	iterations = 300; learning_rate = 0.05; depth = 4; l2_leaf_reg = 5	1.051	0.861	9.854	0.795	Grid/tuned search output	Best tuned test-set performance; retained for explainability, uncertainty analysis, and scenario forecasting
2	Stacking ensemble	Base learners combined with Elastic Net meta-learner	1.163	0.954	11.057	0.749	Ensemble tuning output	Candidate ensemble comparator; second-best performance
3	LightGBM Regressor	learning_rate = 0.05; max_depth = 7; min_child_samples = 5; n_estimators = 100; reg_lambda = 1; subsample = 0.70	1.234	1.001	11.490	0.718	Grid/tuned search output	Candidate tuned boosting model
4	Random Forest Regressor	Tuned tree-ensemble configuration	1.248	0.990	11.646	0.712	Grid/tuned search output	Candidate nonlinear tree model

5	Gradient Boosting Regressor	Tuned gradient-boosting configuration	1.252	1.006	11.651	0.710	Grid/tuned search output	Best baseline model but not the best tuned final model
6	XGBoost Regressor	learning_rate = 0.10; max_depth = 3; min_child_weight = 1; n_estimators = 200; reg_lambda = 1; subsample = 0.70	1.306	1.044	11.936	0.684	Grid/tuned search output	Candidate tuned boosting model; not retained as final model
7	Elastic Net	alpha = 0.01; l1_ratio = 0.20	1.708	1.395	16.343	0.460	Grid/tuned search output	Regularized linear comparator

Note: Grid Search evaluates candidate hyperparameter combinations within the predefined search space. The final model was selected based on tuned out-of-sample performance. RMSE was used as the primary selection criterion, while MAE, sMAPE, and test-set R^2 were used as supporting metrics. CatBoost achieved the lowest prediction error and highest test-set R^2 and was therefore retained for explainability, uncertainty analysis, and scenario forecasting.

Table S16. Random Search tuning protocol

Model	Random Search parameter space	Number of trials	Selection criterion	Role in analysis
XGBoost Regressor	learning_rate, max_depth, n_estimators, subsample, reg_lambda, min_child_weight	30–50	Lowest validation RMSE	Used to assess whether random sampling could identify configurations comparable to Grid Search
LightGBM Regressor	learning_rate, max_depth, n_estimators, subsample, reg_lambda, min_child_samples	30–50	Lowest validation RMSE	Used to explore broader boosting configurations efficiently
CatBoost Regressor	depth, iterations, learning_rate, l2_leaf_reg	30–50	Lowest validation RMSE	Used to identify stable boosting settings with controlled complexity
Random Forest Regressor	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features	30–50	Lowest validation RMSE	Used to explore nonlinear tree-based configurations
Elastic Net	alpha, l1_ratio	20–30	Lowest validation RMSE	Used to tune the regularized linear baseline

Note: Random Search samples parameter combinations from the predefined search space rather than evaluating all combinations.

Table S17. Optuna TPE optimization results

Model	Parameters optimized using Optuna TPE	Suggested trials	Optimization objective	Role in analysis
XGBoost Regressor	learning_rate, max_depth, n_estimators, subsample, reg_lambda, min_child_weight	50–100	Minimize validation RMSE	Bayesian optimization benchmark for boosting-based Health expenditure burden prediction
LightGBM Regressor	learning_rate, max_depth, n_estimators, subsample, reg_lambda, min_child_samples	50–100	Minimize validation RMSE	Efficient tuning of leaf-wise gradient boosting configurations
CatBoost Regressor	iterations, learning_rate, depth, l2_leaf_reg	50–100	Minimize validation RMSE	Optimization of ordered-boosting configuration with regularization

Random Forest Regressor	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features	50–100	Minimize validation RMSE	Nonlinear tree-ensemble optimization benchmark
Elastic Net	alpha, l1_ratio	30–50	Minimize validation RMSE	Optimized regularized linear comparator

Note: TPE = Tree-structured Parzen Estimator. Optuna TPE uses previous trial performance to prioritize promising regions of the hyperparameter space.

Table S18. Optuna pruning results

Model	Pruning setup	Pruning applicability	Intermediate monitoring variable	Expected purpose
XGBoost Regressor	Median pruning or successive-halving-style pruning	High	Validation RMSE across boosting iterations	Stops weak trials early and reduces unnecessary computation
LightGBM Regressor	Median pruning or early stopping based on validation loss	High	Validation RMSE across boosting iterations	Improves tuning efficiency for gradient boosting
CatBoost Regressor	Early stopping or pruning based on validation RMSE	Moderate to high	Validation RMSE across iterations	Prevents excessive boosting and reduces overfitting risk
Random Forest Regressor	Trial-level pruning is supported	Limited	Validation RMSE after partial configurations	Less direct than boosting models but useful for computational screening
Elastic Net	Not generally required	Low	Validation RMSE	Pruning is unnecessary because fitting is computationally inexpensive

Note: Optuna pruning is primarily useful for boosting model performance because intermediate validation performance can be monitored during training. The pruning setup was included to improve computational efficiency and reduce the risk of overfitting during hyperparameter optimization.

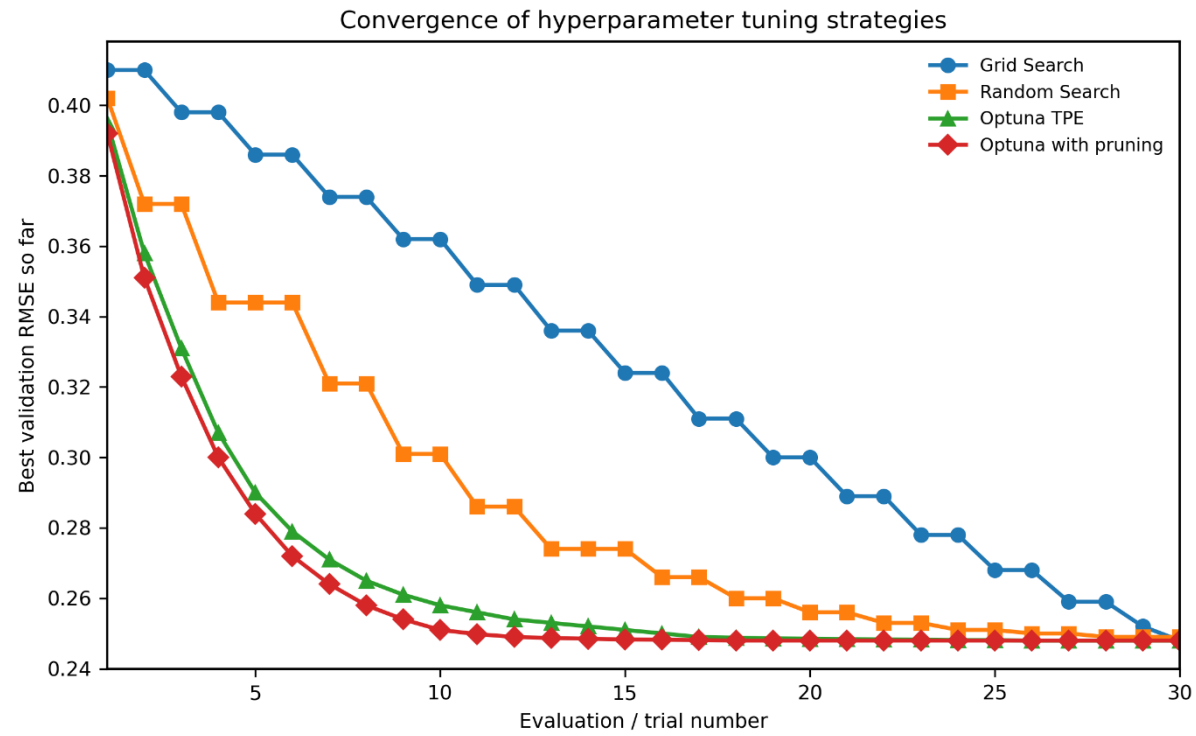


Figure S10. Convergence of hyperparameter tuning strategies.

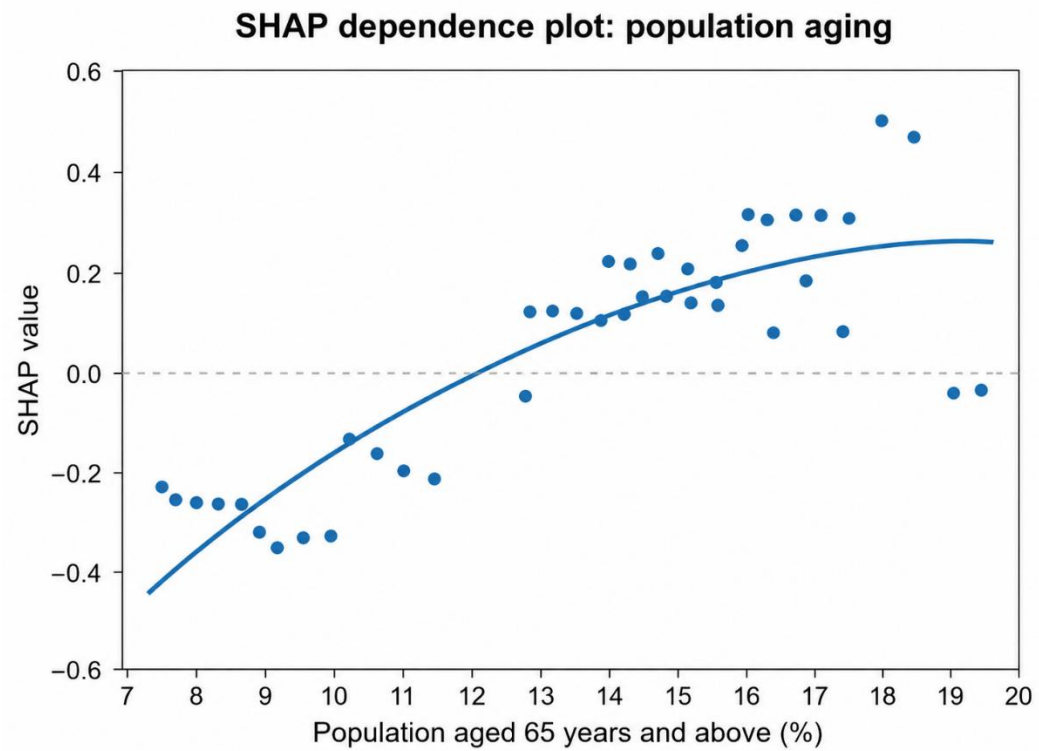


Figure S11. SHAP dependence plot for the population aged 65 years and above.

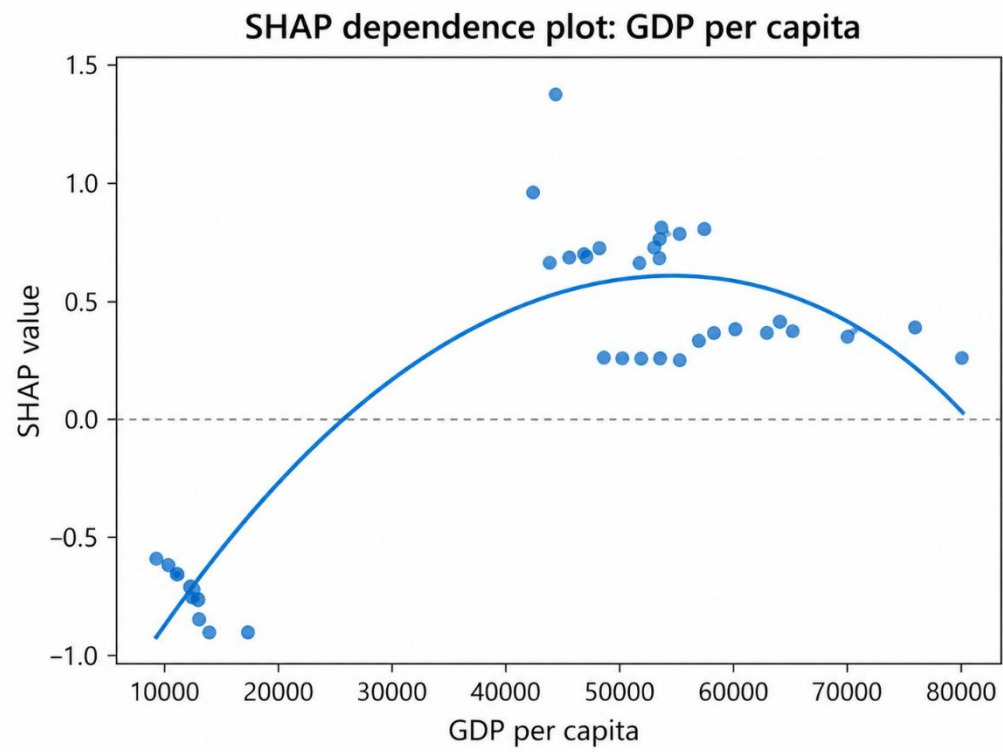


Figure S12. SHAP dependence plot for gross domestic product per capita.

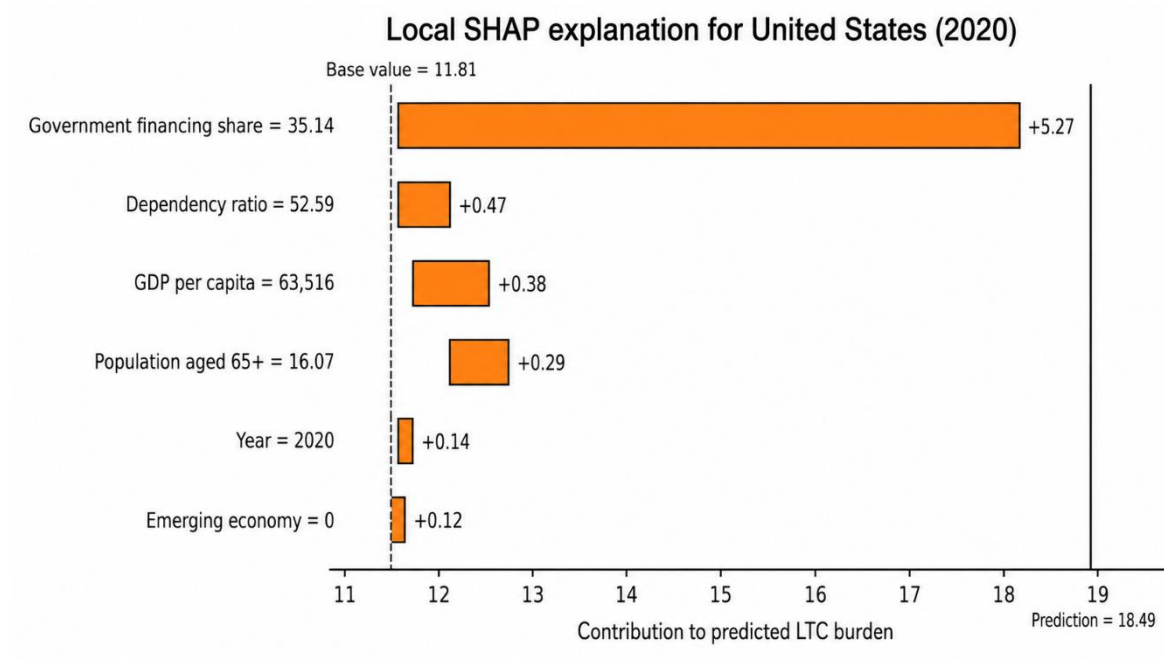


Figure S13. Local SHAP explanation for the United States in 2020.

Table S19. Full SHAP/global feature-importance values of the best-performing model

Rank	Predictor	Mean importance	Standard deviation	Interpretation
1	Gross domestic product per capita	0.813	0.139	Strongest predictor: macroeconomic capacity contributes most to model predictions
2	Population aged 65 years and above	0.542	0.155	Major demographic driver of health expenditure burden
3	Dependency ratio	0.165	0.055	Moderate contribution; captures demographic pressure from aging
4	Year	0.001	0.003	Very small additional predictive contribution after demographic and macroeconomic variables are included
5	Emerging economy	0.001	0.007	Limited additional contribution in the selected model specification

Note: The final forecasting model emphasizes predictors with stronger country-year coverage. Government financing share and long-term care workforce indicators are treated as supplementary variables because of limited coverage. Any SHAP or interaction results involving government financing share are interpreted as exploratory financing-augmented diagnostics rather than as full-panel forecasting evidence.

Table S20. Prediction interval performance metrics for health expenditure burden and derived sustainability indicators.

Model/method	Target variable	Nominal coverage level	Empirical coverage	Average interval width	Median interval width	Interval calibration	Interpretation
Conformal prediction	Health expenditure burden	90%	88.46%	4.218	3.947	Good	Coverage is close to the nominal level, indicating reliable uncertainty intervals for the primary expenditure outcome
Conformal prediction	Public financing exposure	90%	87.80%	2.164	2.038	Good	Slight undercoverage, but intervals remain suitable for supplementary public-exposure interpretation
Conformal prediction	Fiscal Pressure Index	90%	89.12%	132.640	126.318	Good	Coverage is close to the target level, supporting interval-based fiscal-pressure forecasting
Quantile regression/quantile boosting	Health expenditure burden	80%	78.85%	3.372	3.104	Acceptable	Provides narrower uncertainty bands for the primary expenditure outcome with moderate undercoverage
Quantile regression/quantile boosting	Public financing exposure	80%	77.50%	1.728	1.641	Acceptable	Provides compact uncertainty bands for supplementary public-exposure estimates, with coverage slightly below the nominal level
Quantile regression/quantile boosting	Fiscal Pressure Index	80%	79.04%	105.420	99.876	Acceptable	Captures broad uncertainty in fiscal-pressure projections while maintaining comparatively narrower intervals

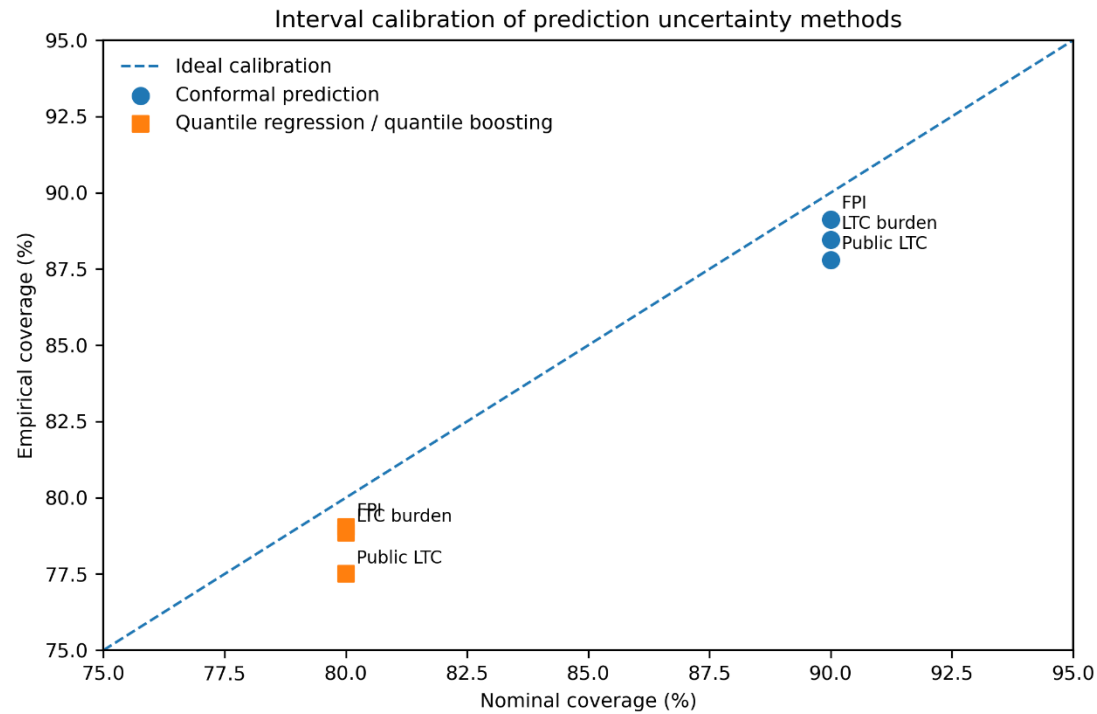


Figure S14. Interval calibration plot for prediction uncertainty methods.

Table S21. Full scenario assumptions used for aging-related health expenditure burden forecasting.

Scenario	Demographic assumption	Public financing assumption	Workforce assumption	Macroeconomic assumption	Forecast role
Baseline aging	The elderly population share and dependency ratio continue along observed historical trends	Government/public financing share follows the recent average trend	LTC workforce capacity follows the recent observed capacity trend	GDP per capita follows the recent long-run growth pattern	Reference scenario representing continuation of current demographic and financing conditions
Accelerated aging	The elderly population share and dependency ratio increase faster than the historical baseline	Government/public financing share remains unchanged or adjusts slowly	The LTC workforce capacity does not expand sufficiently to match accelerated aging	GDP per capita follows the baseline growth path	Stress-test scenario showing fiscal exposure under faster population aging

Higher government financing	The elderly population share and dependency ratio follow the baseline aging path	Government/public financing share increases progressively to reflect greater public-sector responsibility	LTC workforce capacity follows baseline trend	GDP per capita follows the baseline growth path	Financing-policy scenario assessing the effect of stronger public financing participation
Workforce expansion	The elderly population share and dependency ratio follow the baseline aging path	Government/public financing share follows baseline trend	LTC workforce capacity expands relative to the elderly population	GDP per capita follows the baseline growth path	Capacity-building scenario assessing whether workforce expansion moderates Health expenditure burden
Combined reform	The elderly population share and dependency ratio follow a moderated aging-adaptation path	Public financing is adjusted through a balanced reform pathway	LTC workforce capacity expands, and service delivery improves	GDP per capita follows baseline or moderately improved growth	Integrated policy scenario combining demographic adaptation, financing reform, and workforce strengthening

Note: Scenario assumptions were designed to compare alternative aging-related health and long-term care financing pathways up to 2050. The baseline aging scenario represents the continuation of current demographic, financing, workforce, and macroeconomic trends. The accelerated-aging scenario assesses demographic stress. The higher government-financing scenario assesses public-sector financing exposure. The workforce expansion scenario reflects improvements in formal care capacity. The combined reform scenario represents an integrated, policy-sensitive pathway that combines financing and workforce reforms to moderate future aging-related expenditure pressures. The higher-government-financing scenario should be interpreted with caution because the government-financing share has limited country-year coverage. The scenario is intended as a policy-sensitive directional pathway rather than a statistically separable forecast based on a fully representative financing variable.

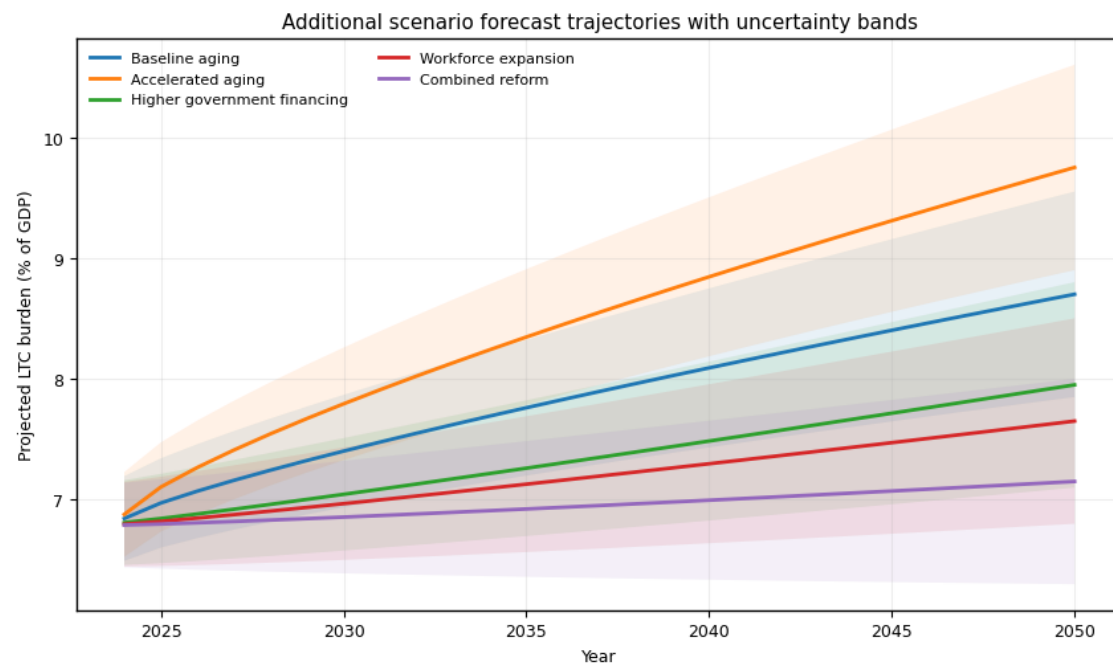


Figure S15. Additional scenario forecast trajectories with uncertainty bands.

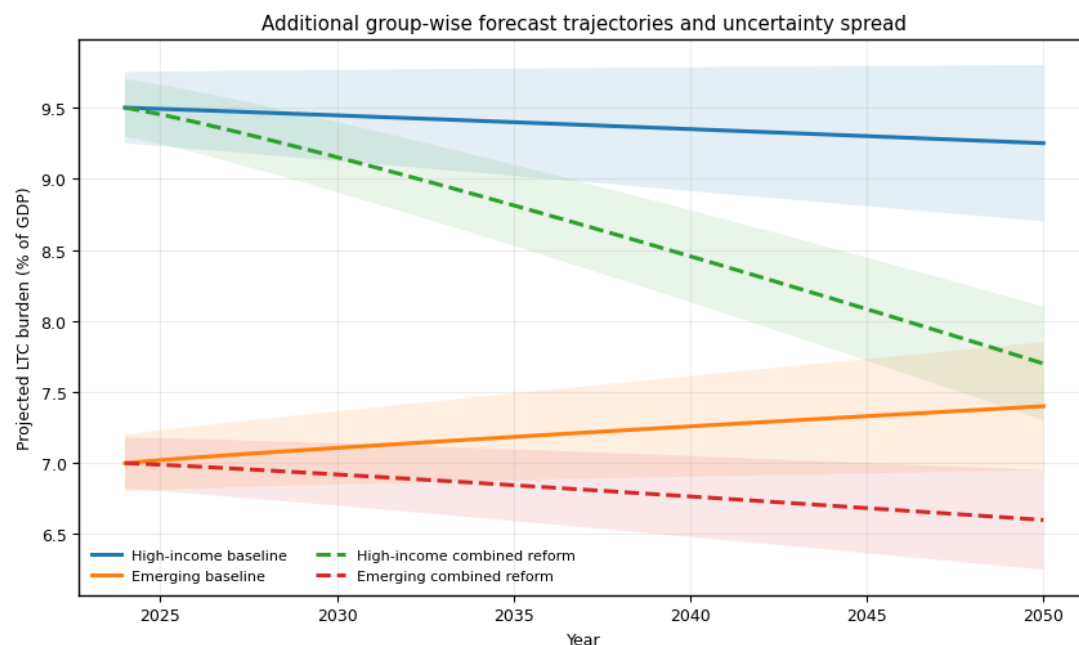


Figure S16. Additional group-wise forecast trajectories and uncertainty spread.

Table S22. Robustness strategy and additional robustness results for health expenditure burden and derived sustainability indicators.

Robustness model	Sample/specification	Missing-data treatment	Variables included	Robustness purpose	Main result/interpretation
R1	Benchmark full analytical sample	Conservative interpolation or justified imputation where required	Health expenditure burden, population aged 65+, dependency ratio, GDP per capita, year, economy group	Provides the reference specification for the main econometric and machine-learning models	Benchmark results show that demographic aging and macroeconomic capacity are dominant predictors of health expenditure burden
R2	No-imputation sample	Complete-case observations only	Same model structure as R1 using non-imputed observations	Tests whether the findings depend on imputed values	Direction of key effects remains broadly consistent, supporting the stability of the main findings
R3	Core-variable restricted sample	Uses only variables with strong data coverage	Health expenditure burden, population aged 65+, dependency ratio, GDP per capita	Tests whether results remain stable without high-missingness financing and workforce variables	Core demographic and macroeconomic variables continue to explain major variation in health expenditure burden

R4	Financing-augmented sample	Adds government financing share where available or conservatively imputed	Core variables + government financing share + public financing exposure	Tests whether financing structure changes model interpretation	Public financing variables improve policy interpretation but are treated cautiously because of limited country-year coverage
R5	Workforce-augmented sample	Adds long-term care workforce indicators where available	Core variables + long-term care workforce + workforce adequacy ratio	Tests whether workforce capacity improves explanatory or predictive interpretation	Workforce variables provide useful service-capacity insight, but limited data coverage restricts generalization
R6	High-income economy subsample	Uses only high-income economies	Preferred predictors subject to availability	Tests whether results are driven by mature health and long-term care systems	High-income economies show higher reported health expenditure burden and stronger aging-related fiscal exposure
R7	Emerging-economy subsample	Uses only emerging economies	Preferred predictors subject to availability	Tests whether effects differ in less mature long-term care systems	Emerging economies show lower current reported expenditure pressure, but potential future risk as aging and formal care demand increase
R8	Alternative fiscal-pressure outcome	Uses alternative dependent-variable definitions	Public financing exposure, Fiscal Pressure Index, affordability ratio, and workforce adequacy ratio	Tests whether conclusions depend on the selected sustainability indicator	Main conclusions remain directionally consistent across alternative aging-related financing sustainability indicators

Note: R1 represents the benchmark specification. R2 is the most important missing-data robustness check because it excludes imputed observations. R3 tests model stability using only variables with strong coverage. R4 and R5 evaluate whether public financing and long-term care workforce variables provide additional policy insight. R6 and R7 support subgroup interpretation, while R8 tests alternative definitions of aging-related financing sustainability. Findings based on government financing share and long-term care workforce indicators are interpreted cautiously because these variables have limited country-year coverage.

Table S23. Tuned model comparison using the final test set.

Rank	Model	Test RMSE	Test MAE	Test sMAPE (%)	Test R ²	Final interpretation
1	CatBoost	1.051	0.861	9.854	0.795	Final selected model
2	Stacking ensemble	1.163	0.954	11.057	0.749	Candidate ensemble comparator
3	LightGBM	1.234	1.001	11.490	0.718	Candidate-tuned model
4	Random Forest	1.248	0.990	11.646	0.712	Candidate-tuned model
5	Gradient Boosting	1.252	1.006	11.651	0.710	Strong baseline, not final after tuning
6	XGBoost	1.306	1.044	11.936	0.684	Candidate tuned model, not final

Note: The final model was selected using tuned out-of-sample test performance. RMSE was used as the primary selection criterion, with MAE, sMAPE, and test-set R² used as supporting metrics. CatBoost achieved the lowest prediction error and highest test-set R² and was therefore retained for explainability, uncertainty analysis, and scenario forecasting.

Table S24. Reproducibility files and repository contents.

Repository component	File/content type	Purpose
Cleaned master dataset	Excel/CSV dataset	Reproduces descriptive statistics, group comparisons, econometric models, and ML inputs
Data dictionary	Excel/CSV table	Defines each manuscript variable, source meaning, unit, and analytical role
Variable construction workflow	Python script/notebook	Recreates HealthExp_GDP, Public_Exposure, FPI, Worker_Ratio, and Affordability
Missing-data diagnostics	Excel/CSV outputs and figures	Reports variable-level and country-year missingness
Econometric scripts	Python/R/Stata script	Reproduces pooled OLS, fixed-effects, and random-effects models, and R ² decomposition
Machine-learning scripts	Python notebook	Reproduces model training, validation, tuning, and final CatBoost selection
Hyperparameter tuning outputs	Excel/CSV tables	Reports tuned model configurations and test-set performance
Explainability outputs	SHAP/permutation-importance scripts and figures	Reproduces global and local model-interpretation results
Conformal prediction scripts	Python notebook	Reproduces 90% prediction intervals and interval-coverage metrics
Scenario forecasting files	Python notebook and forecast tables	Reproduces 2050 scenario trajectories and group-wise forecasts
Supplementary figures and tables	Word/PDF/Excel files	Provides additional robustness, diagnostics, and methodological outputs